Thus $P = 0.3333$ and
$Q = 1 - P = 1 - 0.3333 = 0.6667$.

The S.E for sample proportion is given by

$$S.E. = \sqrt{\frac{PQ}{n}} = \sqrt{\frac{(0.3333)(0.6667)}{9000}} = 0.00496$$

Hence the standard error for sample proportion is S.E=0.00496.

Example 8.9

The standard deviation of a sample of size 50 is 6.3. Determine the standard error whose population standard deviation is 6?

Solution:

Sample size $n = 50$

Sample S.D $s = 6.3$

Population S.D $\sigma = 6$

The standard error for sample S.D is given by

$$S.E. = \sqrt{\frac{\sigma^2}{2n}} = \frac{6}{\sqrt{2(50)}} = \frac{6}{\sqrt{100}} = 0.6$$

Thus standard error for
sample S.D = 0.6.

Example 8.10

A sample of 100 students is chosen from a large group of students. The average height of these students is 162 cm and standard deviation (S.D) is 8 cm. Obtain the standard error for the average height of large group of students of 160 cm?

Solution:

Given $n = 100$, $\bar{x} = 162$ cm, s = 8 cm is known in this problem

since σ is unknown, so we consider $\hat{\sigma} = s$ and φ= 160 cm

$$S.E. = \frac{\hat{\sigma}}{\sqrt{n}} = \frac{s}{\sqrt{n}} = \frac{8}{\sqrt{100}} = 0.8$$

Therefore the standard error for the average height of large group of students of 160 cm is 0.8.

Exercise 8.1

1. What is population?
2. What is sample?
3. What is statistic?
4. Define parameter.
5. What is sampling distribution of a statistic?
6. What is standard error?
7. Explain in detail about simple random sampling with a suitable example.
8. Explain the stratified random sampling with a suitable example.
9. Explain in detail about systematic random sampling with example.
10. Explain in detail about sampling error.
11. Explain in detail about non-sampling error.
12. State any two merits of simple random sampling.
13. State any three merits of stratified random sampling.
14. State any two demerits of systematic random sampling.
15. State any two merits for systematic random sampling.
16. Using the following Tippet's random number table

2952	6641	3992	9792	7969	5911	3170	5624
4167	9524	1545	1396	7203	5356	1300	2693
2670	7483	3408	2762	3563	1089	6913	7991
0560	5246	1112	6107	6008	8125	4233	8776
2754	9143	1405	9025	7002	6111	8816	6446

Draw a sample of 10 three digit numbers which are even numbers.

17. A wholesaler in apples claims that only 4% of the apples supplied by him are defective. A random sample of 600 apples contained 36 defective apples. Calculate the standard error concerning of good apples.
18. A sample of 1000 students whose mean weight is 119 lbs(pounds) from a school

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